

Relaytic-AML: Claim-Gated Evaluation Environments for Temporal Graph Financial-Crime ML

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Abstract

Financial-crime machine learning is difficult to publish responsibly because the same metric can mean different things under different temporal splits, graph provenance, review-capacity assumptions, dataset realism, and public claim scopes. Relaytic-AML is a local-first evaluation environment that treats those constraints as auditable first-class artifacts. The central unit is an evidence cell: a metric bound to a dataset registry entry, split contract, execution command, artifact field, budget tier, leakage posture, operating-point policy, and publishability gate. A claim gate then maps those cells into supporting-only, blocked, or headline-eligible evidence before paper text or public wording can use them. In the current release pack, a competitive PaySim synthetic temporal-fraud row reports test PR-AUC 0.638773, and an Elliptic temporal graph-feature row reports test PR-AUC 0.668756; both are supporting evidence only. Elliptic2 subgraph evidence is retained as modern context only: the local repeated official-partition candidate reports PR-AUC 0.94324 +/- 0.000882, below the recorded RevClassifyDS reference of 0.974, and reference-parity plus cohort gates remain unresolved. The contribution is therefore not a detector superiority claim. It is a reproducible AML evaluation environment for preventing benchmark scores, operational estimates, limitations, and public claims from drifting apart.

1 Introduction

AML systems are operational decision systems, not just classifiers. Investigators need temporally valid predictions, graph or entity provenance, a calibrated review queue, and defensible statements about what the evidence does and does not prove. Public AML benchmarks make this harder because datasets vary across synthetic mobile-money simulation, transaction-level graph labels, and modern subgraph labels. A result that is useful on one track can be misleading if it is promoted as a broader real-world AML claim.

Modern AML work is moving quickly toward large subgraph benchmarks, profile-aware simulation, continual graph learning, and high-throughput graph mining. Those directions are valuable, but they make evidence management harder: an agent or researcher can produce plausible tables while quietly mixing proxy datasets, exposed partitions, unfrozen thresholds, or unsupported business claims. Relaytic-AML addresses that reliability problem by turning the evaluation process itself into the object under test.

The system records benchmark inputs, split rules, model-search budgets, leakage posture, operating points, review-budget estimates, figure provenance, and public-claim gates. The result is a paper package where the reader can inspect exactly why a row is allowed as supporting evidence and why stronger wording remains blocked.

2 Contributions

This paper makes five contributions.

1. An evidence-cell contract for AML benchmark rows that binds each reported number to dataset, split, command, artifact field, budget tier, leakage posture, operating-point policy, and claim state.
2. A deterministic claim-gate layer that separates supporting evidence from blocked or headline-eligible claims before paper text and public wording are generated.
3. A budget ladder that distinguishes smoke, baseline, competitive, and release runs so weak first-pass rows cannot be laundered into strong claims.
4. A release-safety pipeline that blocks public wording when clean-clone, claim-lint, leak-scan, citation, figure, or publishability gates fail.
5. A transparent first evidence pack over PaySim, Elliptic, and Elliptic2-context tracks that preserves limitations instead of converting proxy or blocked evidence into broader claims.

3 Research Questions

The paper evaluates three systems questions.

1. Can AML benchmark evidence be represented as reproducible local artifacts rather than as prose-level claims?
2. Can claim gates prevent supporting proxy, graph-feature, and modern-context evidence from becoming unsupported hard AML or benchmark-superiority claims?
3. Can the same artifact contract help humans, external agents, and future research automation understand what is proven, what is blocked, and what work would unlock stronger claims?

4 Related Work

PaySim is a synthetic mobile-money simulator designed to address the scarcity of legitimate public mobile-transaction datasets for fraud research Lopez-Rojas et al. [2016]. It is useful for temporal transaction-fraud workflow evaluation, but synthetic evidence cannot by itself establish hard real-bank AML performance.

The Elliptic Bitcoin dataset introduced a public transaction graph with more than 200K transaction nodes, 234K directed payment-flow edges, and 166 node features across 49 time steps Weber et al. [2019]. That work also showed why graph evidence must be compared against strong simpler baselines rather than assumed superior.

Elliptic2 shifts the public AML benchmark center toward subgraph learning, with 121,810 labeled subgraphs inside a background graph of roughly 49M node clusters and 196M edge transactions Bellei et al. [2024]. RevTrack and RevClassify further argue that sender and receiver context around a subgraph can be a powerful and scalable signal Song et al. [2024]. These

works motivate Relaytic-AML’s modern-context and limitation track, but they do not make the current Relaytic Elliptic2 row a performance contribution.

The 2025/2026 AML graph literature also raises the bar beyond the current Relaytic evidence rows. TransXion frames benchmark realism around profile-aware simulation and out-of-character behavior Chen et al. [2026]. LineMVGNN and ExSTraQt represent detector-focused work on directed money flow, edge-aware graph views, and quasi-temporal transaction representations Poon et al. [2026], Tariq and Hassani [2026]. BlazingAML stresses throughput and fuzzy multi-stage scheme expression as a systems problem Ye et al. [2026], while continual graph-learning reviews emphasize drift, adaptation, class imbalance, and evolving laundering behavior Deprez et al. [2025]. Relaytic-AML is positioned as complementary infrastructure for such work: it does not claim detector parity with these systems, but it makes dataset posture, split validity, budgets, limitations, and public claims auditable.

The paper also follows broader ML documentation and reproducibility practice. Datasheets for Datasets and Model Cards argue for explicit dataset and model reporting Gebru et al. [2021], Mitchell et al. [2019]. The NeurIPS reproducibility program highlights the need for code, data, and checklist discipline in ML research Pineau et al. [2021]. Recent work on ML research agents warns that coherent papers can still contain invalidated experiments, reinforcing the need for executable artifacts and claim gates Chen et al. [2025].

5 Method: Evidence Cells and Claim Gates

Relaytic-AML treats a paper metric as an evidence cell, not as a free-standing score. An evidence cell is a tuple consisting of dataset identity, split contract, execution command, run-directory or artifact reference, artifact field, metric value, budget tier, leakage posture, operating-point rule, claim state, and gate limitation notes. The table generator reads only these cells, and the paper generator cites the cell identifier for every numeric row.

A claim gate is a deterministic predicate over evidence cells and release artifacts. In this release, the gate can assign a row to supporting-only evidence, modern-context-only evidence, baseline-only evidence, or blocked evidence. No row is headline-eligible. The same gate also lints generated paper text and public wording, so unsupported phrases such as hard AML superiority, RevClassify parity, graph-neural superiority, hard business value, or leaderboard-winner claims remain blocked.

This method is intentionally conservative. It is designed for settings where a strong-looking model number can be less important than the question of whether the number was produced under a valid split, compared against an appropriate baseline, selected without test leakage, and described with the right scope.

6 Relaytic-AML Evaluation Environment

Relaytic-AML is organized as a deterministic local evidence pipeline. Dataset registry artifacts define source and access posture. Split contracts define chronological, graph-snapshot, or subgraph partition rules. Benchmark runners produce model and operating-point artifacts. Paper-table generation consumes those artifacts and writes per-cell provenance. Draft, release, and arXiv-source generation then lint wording against publishability gates.

The environment has three design rules.

1. Local artifacts are the source of truth. Narrative text is derived from artifacts, not the reverse.
2. Validation selects models, thresholds, and operating points before fixed test evaluation.
3. Blocked evidence stays visible as a limitation, because hiding failed or incomplete tracks makes both human and agentic research less scientific.

Figure 1. Claim-gated evidence flow (schematic)



Local artifacts remain the source of truth; unsupported benchmark tracks stay visible as blocked limitations.

Figure 1: Relaytic-AML claim-gated evidence flow

Figure 2. Supporting PR-AUC evidence (not headline claims)

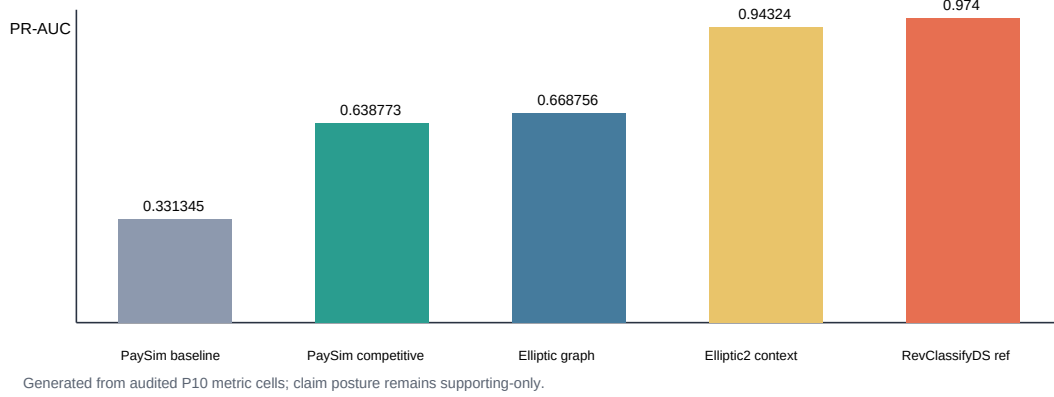


Figure 2: Supporting PR-AUC rows with claim posture

7 Benchmark Protocol

The current release pack separates smoke, baseline, competitive, and release budgets. Smoke checks prove that commands and artifacts exist. Baseline budgets establish conservative full-dataset evidence where possible. Competitive budgets use stronger features, candidate families, calibration, and validation-only operating-point selection. Release budgets freeze the paper transformation path and require clean-clone and leak-scan proof. A budget tier is part of the evidence cell, so the paper can distinguish a serious competitive run from a quick reproducibility check.

PaySim is treated as a synthetic temporal proxy. Elliptic is treated as temporal graph-feature supporting evidence. Elliptic2 is treated as modern subgraph context and limitation evidence, because the current local environment has not executed faithful RevClassify parity and the current-core to RevTrack-evaluable cohort boundary is not fully proven.

Figure 3. Review-budget precision and recall

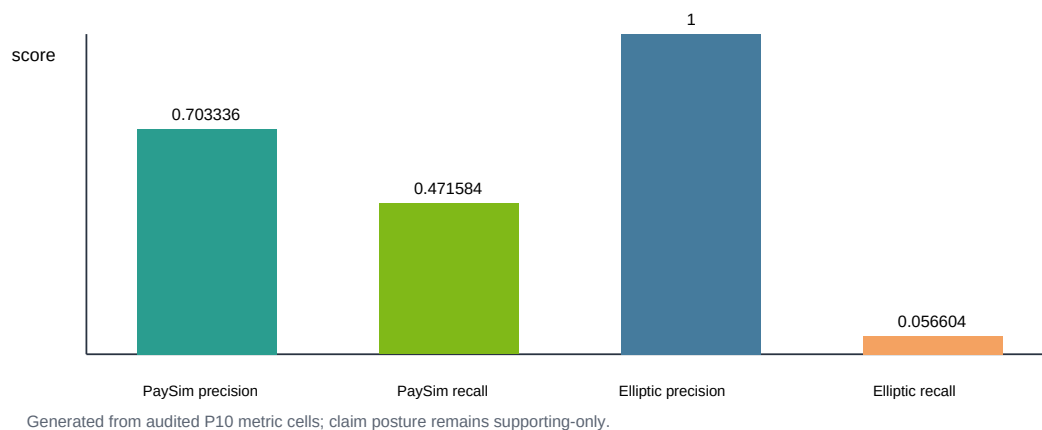


Figure 3: Review-budget precision and recall

Figure 4. Publishability gate posture by track

	Supporting	Headline	Hard claim	Gate
paysim_temporal_transaction_fraud	yes	no	no	pass_supporting_only
elliptic_flattened_graph_aml	yes	no	no	pass_supporting_only
elliptic2_subgraph_aml	yes	no	no	pass_supporting_modern_context_or
elliptic2_subgraph_aml	yes	no	no	blocked_supporting_only_thesis_nan
paper_operational_layer	yes	no	no	supporting_operational_metrics_read

Figure 4: Publishability gate posture by track

Evidence row	Metric	Value	Claim posture	Provenance
PaySim baseline	test PR-AUC	0.331345	baseline-only	paper-cell:paysim_p6_valid
PaySim	test PR-AUC	0.638773	supporting-only	paper-cell:paysim_p6a_comp
competitive			synthetic	
			temporal proxy	
PaySim	precision at	0.703336	supporting-only	paper-cell:paysim_p6a_comp
competitive	review budget			
PaySim	recall at review	0.471584	supporting-only	paper-cell:paysim_p6a_comp
competitive	budget			
Elliptic	test PR-AUC	0.668756	supporting-only	paper-cell:elliptic_p7_sel
graph-feature			graph-feature	
			evidence	
Elliptic	precision at	1	supporting-only	paper-cell:elliptic_p7_sel
graph-feature	review budget			
Elliptic	recall at review	0.056604	supporting-only	paper-cell:elliptic_p7_sel
graph-feature	budget			
Elliptic2 context	official-partition	0.94324	modern context	paper-cell:elliptic2_p8b_m
	PR-AUC mean		only	
Elliptic2 context	official-partition	0.000882	modern context	paper-cell:elliptic2_p8b_m
	PR-AUC std		only	
RevClassifyDS	published	0.974	reference	paper-cell:elliptic2_p8b_m
reference	PR-AUC		context, not	
			parity	

All values are generated from `docs/reports/paper_metric_cell_audit.json`; none is a headline or hard AML claim.

8 Results

The PaySim competitive row improves over the PaySim baseline inside the recorded synthetic temporal-fraud contract. The Elliptic graph-feature row is supporting graph evidence with modest structural lift, not graph-neural superiority. The Elliptic2 context row is strong enough to motivate future reprovisioning, but not enough to claim parity with the RevClassifyDS reference or to make an Elliptic2 performance contribution.

The most important result is the gate outcome: all current numeric rows are usable for a claim-safe evaluation-environment paper, and none is allowed to become a hard AML, production, benchmark-superiority, or business-value claim. That outcome is scientifically useful because it exposes what the environment knows and what it refuses to overstate.

Track	Supporting table	Headline claim	Hard claim	Gate status	Gate limitation notes
paysim_temporal	yes	transaction_fraud	no	pass_supporting	paysim_is_supporting_pro re- lease_budget_and_clean_cl

Track	Supporting table	Headline claim	Hard claim	Gate status	Gate limitation notes
elliptic_flattened_graph_aml	yes	no	no	pass_supporting	only headline_graph_claim_req graph_sota_claim_not_ben hard_aml_claim_requires_ graph_neural_shadow_can
elliptic2_subgraph_aml	yes	no	no	pass_supporting	official test partition was en- tity_disjoint_generalizatio
elliptic2_subgraph_aml	yes	no	no	blocked_supporting	fig only throws mapping local accelerator not availa offi- cial_revclassify_classificat offi- cial_revclassify_dependen core_missing, offi- cial_revclassify_dependen offi- cial_revclassify_dependen +4 more
paper_operational_klayer	yes	no	no	supporting_operational	paper metrics already sha ana- lyst_hour_assumption_def same_queue_incumbent_o opera- tional_metrics_supporting ellip- tic2_excluded_from_opera

9 Discussion

The practical value of Relaytic-AML is not that it replaces a compliance platform. Its value is that it can give risk, fraud, or AML teams a local evidence lab for unknown datasets, incumbent challenges, review-budget tradeoffs, leakage checks, and public-claim discipline. A company evaluating a new dataset can inspect whether the result is a model win, an environment win, a proxy-only result, or a blocked claim. That distinction is often what separates a useful internal experiment from an unsafe public or deployment claim.

For research teams and agentic ML workflows, the same structure is a guard against coherent but invalid experiments. External agents can consume the generated JSON artifacts, reproduce the paper tables, see blocked claims, and propose the next benchmark action without inferring hidden state from prose. The strongest story is therefore the artifact discipline: Relaytic-AML demonstrates that an ambitious evaluation system can still refuse claims it has not earned.

10 Limitations

- **LIM-01-paysim-proxy:** PaySim is synthetic mobile-money fraud evidence. It is useful for a temporal proxy workflow, but it is not real-bank AML superiority evidence. Required repair: Add a real financial-crime holdout or partner-approved private evaluation before making hard AML claims.
- **LIM-02-elliptic-supporting-graph:** The Elliptic row is a supporting temporal graph-feature result. It does not prove graph-neural or graph-SOTA superiority. Required repair: Run repeated-seed graph baselines and promote a graph-native candidate only if it beats strong feature baselines under the same split.
- **LIM-03-elliptic2-context-only:** Elliptic2 is retained as modern context and limitation evidence only; it is not a Relaytic performance contribution in this paper. Required repair: Reproduce a faithful RevClassify parity run or define a new leakage-resistant subgraph protocol with viable cohort proof.
- **LIM-04-operational-assumptions:** Operational review-budget rows are supporting estimates because aggregate case packets, same-queue incumbent comparisons, and analyst-hour assumptions are not fully frozen. Required repair: Freeze case-packet completeness and compare against the same review queue or an approved incumbent baseline.
- **LIM-05-clean-clone-smoke-scope:** P12 clean-clone and paper-smoke proof now passes for the generated paper path, including install-readiness checks, P10/P11 smoke regeneration, leak scan, and reproduction failure reporting. The remaining limitation is scope: heavy external-local benchmark reruns are documented but not rerun inside the P12 smoke proof. Required repair: run the heavy benchmark commands under a frozen release budget before promoting hard or headline benchmark claims.

11 Reproducibility

The P13/P14 package is generated from P10-P12 evidence artifacts. The clean-clone proof records install readiness, paper-smoke regeneration, claim lint, leak scan, and failure reporting. The final public wording is constrained by `docs/reports/paper_public_claims_allowed.json`, and the arXiv source bundle is audited by `docs/reports/paper_submission_package_audit.json`.

Artifact	Present	Role
<code>docs/reports/paper_result_table_final.json</code>	gate	P10-P12 gate input
<code>docs/reports/paper_metric_claim_audit.json</code>	gate	P10-P12 gate input
<code>docs/reports/paper_publishability_matrix.json</code>	gate	P10-P12 gate input
<code>docs/reports/paper_claim_lint_report.json</code>	gate	P10-P12 gate input
<code>docs/reports/paper_external_dry_run_report.json</code>	gate	P10-P12 gate input
<code>docs/reports/paper_reproduction_failure_report.json</code>	gate	P10-P12 gate input
<code>docs/reports/paper_release_gosno-go.json</code>	gate	P10-P12 gate input
<code>docs/paper/relaytic_aml_drafts.md</code>	draft	paper draft or figure input
<code>docs/paper/figures/figure_manifest.json</code>	draft	paper draft or figure input
<code>docs/paper/arxiv_src/figures/figure_1_claim_gate_flow.pdf</code>	draft	paper draft or figure input
<code>docs/paper/arxiv_src/figures/figure_2_supporting_pr_audit.pdf</code>	draft	paper draft or figure input
<code>docs/paper/arxiv_src/figures/figure_3_review_budget.pdf</code>	draft	paper draft or figure input
<code>docs/paper/arxiv_src/figures/figure_4_publishability_matrix.pdf</code>	draft	paper draft or figure input

Core reproduction commands:

11.1 Paper P10 Reproduction Commands

Run from the repository root. External-local dataset paths are intentionally placeholders when the source is not committed.

```
relaytic release-safety paysim-benchmark --format json
relaytic release-safety elliptic-graph --format json
relaytic release-safety tabular-baselines --budget-tier baseline --run-optional --format json
relaytic release-safety paysim-competitive --budget-tier competitive --run-optional --format json
relaytic release-safety graph-baselines --budget-tier competitive --run-optional --format json
relaytic release-safety hard-graph-tracks --format json
relaytic release-safety elliptic2-recovery --core-data-dir <external-local-core-dir> --revtrack-dir <external-local-revtrack-dir>
relaytic release-safety elliptic2-competitive --revtrack-dir <external-local-revtrack-dir>
relaytic release-safety elliptic2-reference-parity --revtrack-dir <external-local-revtrack-dir>
relaytic release-safety paper-thesis-decision --format json
relaytic release-safety paper-operational-metrics --format json
relaytic release-safety paper-tables --format json
```

- Table status: `tables_generated_claim_guarded`
- Metric audit status: `pass`
- P9 dependency: `supporting_operational_metrics_ready_hard_claims_blocked`
- Paper may continue to P11: `True`

P13 release command:

```
relaytic release-safety paper-release --format json
relaytic release-safety paper-arxiv-source --format json
relaytic scan-git-safety
```

12 Conclusion

Relaytic-AML should be read as a claim-gated AML evaluation-environment paper. The current evidence pack is useful and publishable in that systems sense: it has real numeric supporting rows, modern benchmark context, deterministic figures, limitations, clean-clone proof, arXiv source packaging, and public wording gates. The same evidence does not support a hard AML superiority, headline benchmark, graph-neural superiority, RevClassify parity, or hard business-value claim. That restraint is part of the contribution.

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